

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Third Semester of B. Tech (CE) Examination****March-April 2018****CE216.02/CE216 Computer Organization and Peripherals****Date: 28.03.2018, Wednesday****Time: 01.30 pm To 04.30 pm****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 Answer the question below. [07]**
- a. Perform following arithmetic operation using 2's complement integers. [1]
i). $35+(-10)$ ii). $20-(-4)$
 - b. What is instruction pipeline? [1]
 - c. What are the flags available in basic computer? [1]
 - d. Write down the classification of computer according to Flynn. [1]
 - e. Draw Only: Flow chart of Booth Multiplication. [3]

- Q – 2.a Convert the following arithmetic expression to postfix notation [04]**
- (1) $A * B * (C + D * D) / F$
 - (2) $A * B * (C + (D / F)) / K$
 - (3) $A + B * C / (G * H) / K$
 - (4) $A + B / C$

OR

- Q – 2.a What are the problems associated with Instruction pipeline? [04]**

- Q – 2.b Answer any TWO. [10]**
- (i) How does CPU manipulate stack using Stack Pointer (SP)?
 - (ii) Write down the characteristics of CISC and RISC architecture.
 - (iii) What are the basic difference between branch instruction, Subroutine call and Interrupt?

- Q – 3. Answer any TWO. [14]**
- a. By taking suitable example, explain various addressing modes.
 - b. Draw and explain block diagram of General Register Organization of Computer. How the control word is crated for $R1 \leftarrow R2 + R3$ Operation.
 - c. Write a short note on Design of control Unit.

SECTION – II**Q - 4 Answer the question below.****[07]**

- What is register transfer language?
- What are the three states of Three-State Buffers?
- Draw Only: Digital Circuit of Binary Incrementer.
- How many memory references are required in direct address and indirect address?
- What are the four phases of instruction execution?
- Explain following terms:
Binary Code, Symbolic Code
- What is pseudo instruction?

Q – 5.a Provide the significance of following registers in CPU.
PC, AR, DR, IR, INPR, OUTR, AC, TR

[04]**Q – 5.b Answer any two questions.****[10]**

- List and explain Shift micro-operations.
- How does microprocessor handle interrupts. Draw Flow chart.
- By taking suitable subroutine assembly language program explain, What is subroutine?

Q - 6 Answer any TWO.**[14]**

- Draw Common Bus System diagram and explain procedure of transferring data from memory M using address location AR to register AC.
- Draw Arithmetic circuit for following function table.

Select			Input Y	Output $D = A + Y + C_{in}$	Microoperation
S_1	S_0	C_{in}			
0	0	0	B	$D = A + B$	Add
0	0	1	B	$D = A + B + 1$	Add with carry
0	1	0	$\overline{B}$	$D = A + \overline{B}$	Subtract with borrow
0	1	1	$\overline{B}$	$D = A + \overline{B} + 1$	Subtract
1	0	0	0	$D = A$	Transfer A
1	0	1	0	$D = A + 1$	Increment A
1	1	0	1	$D = A - 1$	Decrement A
1	1	1	1	$D = A$	Transfer A

- Write an assembly language program to multiply two positive numbers. (Numbers are 13_{10} , 10_{10})
